STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-SMOOTHING-PROPER-CURVE-ISOLATED-LCI

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3328-3340 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 3342-3413, the complete author proof (70 body lines).

Editorial note. Navigation stubs: 11 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 1 of queue_examples-defos.json (pb=3364, stmt 3328-3340, proof 3342-3413).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00SB: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3413–3413. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06HN: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3413–3413. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E7T: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3117–3124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DZP: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3031–3037. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E7W: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3261–3266. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E7X: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3293–3299. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07ZA: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3413–3413. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D4G: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3413–3413. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02NM: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3413–3413. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E7R: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3413–3413. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06LH: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3413–3413. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let X be a scheme over a field k. Assume*

(1) *X is proper over k,*
(2) *X is a local complete intersection over k,*
(3) *X has dimension ≤ 1, and*
(4) *$X \to \mathrm{Spec}(k)$ is smooth except at finitely many points.*

Then there exists a flat projective morphism $Y \to \mathrm{Spec}(k[[t]])$ whose generic fibre is smooth and whose special fibre is isomorphic to X.

Proof. Observe that X is Cohen-Macaulay, see Algebra, Lemma 00SB. Thus $X = X' \amalg X''$ with $\dim(X') = 0$ and X'' equidimensional of dimension 1, see Morphisms, Lemma 02NM. Since X' is finite over k (Varieties, Lemma 06LH) we can find $Y' \to \mathrm{Spec}(k[[t]])$ with special fibre X' and generic fibre smooth by Lemma 0E7W. Thus it suffices to prove the lemma for X''. After replacing X by X'' we have X is Cohen-Macaulay and equidimensional of dimension 1.

We are going to use deformation theory for the situation $\Lambda = k \to k$. Let $p_1, \ldots, p_r \in X$ be the closed singular points of X, i.e., the points where $X \to \mathrm{Spec}(k)$ isn't smooth. For each i we pick an integer n_i and a flat, essentially of finite type ring map

$$k[[t]] \longrightarrow B_i$$

with $B_i/tB_i \cong \mathcal{O}_{X,p_i}$ such that t^{n_i} is in the 1st Fitting ideal of $\Omega_{B_i/k[[t]]}$. This is possible by Lemma 0E7X. Observe that the system (B_i/t^nB_i) defines a formal object of $\mathcal{D}ef_{\mathcal{O}_{X,p_i}}$ over $k[[t]]$. By Lemma 0DZP the map

$$\mathcal{D}ef_X \longrightarrow \prod\nolimits_{i=1,\ldots,r} \mathcal{D}ef_{\mathcal{O}_{X,p_i}}$$

is a smooth map between deformation categories. Hence by Formal Deformation Theory, Lemma 06HN there exists a formal object (X_n) in $\mathcal{D}ef_X$ mapping to the formal object $\prod_i (B_i/t^n)$ by the arrow above. By More on Morphisms of Spaces,

Lemma 0E7R there exists a projective scheme Y over $k[[t]]$ and compatible isomorphisms $Y \times_{\mathrm{Spec}(k[[t]])} \mathrm{Spec}(k[t]/(t^n)) \cong X_n$. By More on Morphisms, Lemma 0D4G we see that $Y \to \mathrm{Spec}(k[[t]])$ is flat. Since X is Cohen-Macaulay and equidimensional of dimension 1 we may apply Lemma 0E7T to check Y has smooth generic fibre[1]. Choose n strictly larger than the maximum of the integers n_i found above. If we can show t^{n-1} is in the first Fitting ideal of Ω_{X_n/S_n} with $S_n = \mathrm{Spec}(k[t]/(t^n))$, then the proof is done. To do this it suffices to prove this is true in each of the local rings of X_n at closed points p. However, if p corresponds to a smooth point for $X \to \mathrm{Spec}(k)$, then $\Omega_{X_n/S_n,p}$ is free of rank 1 and the first Fitting ideal is equal to the local ring. If $p = p_i$ for some i, then

$$\Omega_{X_n/S_n,p_i} = \Omega_{(B_i/t^nB_i)/(k[t]/(t^n))} = \Omega_{B_i/k[[t]]}/t^n\Omega_{B_i/k[[t]]}$$

Since taking Fitting ideals commutes with base change (with already used this but in this algebraic setting it follows from More on Algebra, Lemma 07ZA), and since $n - 1 \geq n_i$ we see that t^{n-1} is in the Fitting ideal of this module over B_i/t^nB_i as desired. □

[1]Warning: in general it is **not** true that the local ring of Y at the point p_i is isomorphic to B_i. We only know that this is true after dividing by t^n on both sides!